PCY & Apriori Practice Numerical Problem Set

Problem 1: PCY Algorithm (Alphabet Items)

A retail store records 4 transactions containing 5 unique items (a, b, c, d, e).

Transactions:

T1 = {a, b, c, e}
T2 = {a, b, d}
T3 = {b, c, e}
T4 = {a, c, d, e}

Parameters:

- Minimum Support Threshold (s): 2 transactions
- Hash Table Size (k): 5 buckets (indexed 0 to 4)
- Hash Function: h (x, y) = (val(x) + val(y)) mod 5, where val(a) = 1, val(b) = 2, val(c) = 3, val(d) = 4, val(e) = 5.

Tasks:

1. Calculate individual item counts and list L1.
2. Hash all 2-item pairs in each transaction to construct the bucket count array and resulting Bit Vector.
3. Generate candidate pairs (C2) using PCY bit-vector filtering.
4. Compute actual counts for C2 and list frequent 2-itemsets (L2).

Problem 2: Standard Apriori Algorithm

A small grocery database contains 5 transactions across items (a, b, c, d).

Transactions: T1 = {a, b, c}, T2 = {a, c}, T3 = {b, c, d}, T4 = {a, b, c, d}, T5 = {a, d}
Parameters: Minimum Support Threshold (s): 3 transactions (60%)

Tasks:

1. Find L1 by computing support counts for singletons.
2. Form Candidate 2-itemsets (C2) using standard Apriori join (L1 x L1) and calculate L2.
3. Form Candidate 3-itemsets (C3) using Apriori pruning rules, verify support, and find L3.

Problem 3: PCY Hash Collision Analysis

An e-commerce dataset yields L1 = {a, b, c, d}. The hash table size is k = 4 buckets (0 to 3), and s = 3.

The hash function is h(x, y) = (val(x) × val(y)) mod 4 with a=1, b=2, c=3, d=4.

During Pass 1, the dataset yields the following 2-item pair counts:

Pair Counts: {a, b} = 2, {a, c} = 1, {a, d} = 1, {b, c} = 2, {b, d} = 3, {c, d} = 2

Tasks:

1. Map each pair to its bucket index and calculate total counts for Buckets 0, 1, 2, 3.
2. Construct the Bit Vector.
3. Identify which candidate pair in C2 is a False Positive (pruned pair that passes the bit vector check due to collision).

Answer Key — Problem 1: PCY Algorithm

1. Pass 1 Item Counts & L1:

Counts: a: 3, b: 3, c: 3, d: 2, e: 3
Result: L1 = {{a}, {b}, {c}, {d}, {e}} (All items meet s = 2)

2. Bucket Hash Mapping:

T1 = {a, b, c, e}: {a, b} → 3 mod 5 = 3; {a, c} → 4; {a, e} → 1; {b, c} → 0; {b, e} → 2; {c, e} → 3
T2 = {a, b, d}: {a, b} → 3; {a, d} → 0; {b, d} → 1
T3 = {b, c, e}: {b, c} → 0; {b, e} → 2; {c, e} → 3
T4 = {a, c, d, e}: {a, c} → 4; {a, d} → 0; {a, e} → 1; {c, d} → 2; {c, e} → 3; {d, e} → 4

3. Bucket Aggregation & Bit Vector (s = 2):

Bucket	Hashed Pairs & Individual Counts	Total Count	Bit Vector (≥ 2)
0	{b,c} (2) + {a,d} (1)	3	1
1	{a,e} (2) + {b,d} (1)	3	1
2	{b,e} (2) + {c,d} (1)	3	1
3	{a,b} (2) + {c,e} (3)	5	1
4	{a,c} (2) + {d,e} (1)	3	1

Bit Vector Result
Bit Vector = [1, 1, 1, 1, 1]
Since all buckets reach support count ≥ 2, no candidate pairs are pruned by the bit vector in this scenario.

4. PCY Candidate Set (C2) & L2 Verification:

C2 Pairs: All 10 combinations of L1 pairs are retained in C2.

Actual Counts: {a, b}: 2, {a, c}: 2, {a, d}: 1, {a, e}: 2, {b, c}: 2, {b, d}: 1, {b, e}: 2, {c, d}: 1, {c, e}: 3, {d, e}: 1
Frequent 2-Itemsets (L2): L2 = {{a, b}, {a, c}, {a, e}, {b, c}, {b, e}, {c, e}}

Problem 2: Standard Apriori Algorithm

1. Pass 1 (L1):

Counts: a: 4, b: 3, c: 4, d: 3
Result: L1 = { {a}, {b}, {c}, {d} }

2. Pass 2 (C2 & L2):

Candidates (C2): C2 = {{a,b}, {a,c}, {a,d}, {b,c}, {b,d}, {c,d}}
Actual Counts: {a,b}: 2, {a,c}: 3, {a,d}: 2, {b,c}: 3, {b,d}: 2, {c,d}: 2
Frequent 2-Itemsets (L2): L2 = {{a,c}, {b,c}}

3. Pass 3 (C3 & L3):

Pruning: To form a 3-itemset {a, b, c}, all 2-item subsets ({a, b}, {a, c}, {b, c}) must be in L2. Since {a, b} $\notin$ L2, {a,b,c} is pruned.
Result: C3 = $\emptyset$, L3 = $\emptyset$

Answer Key — Problem 3: PCY Hash Collision Analysis

1. Hash Buckets h(x, y) = (val(x) × val(y)) mod 4:

- {a,b} → (1×2) mod 4 = 2 (Count = 2)
- {a,c} → (1×3) mod 4 = 3 (Count = 1)
- {a,d} → (1×4) mod 4 = 0 (Count = 1)
- {b,c} → (2×3) mod 4 = 2 (Count = 2)
- {b,d} → (2×4) mod 4 = 0 (Count = 3)
- {c,d} → (3×4) mod 4 = 0 (Count = 2)

2. Bucket Totals & Bit Vector (s = 3):

Bucket	Hashed Pairs & Counts	Total Count	Bit Vector (≥ 3)
0	{a,d} (1) + {b,d} (3) + {c,d} (2)	6	1
1	None	0	0
2	{a,b} (2) + {b,c} (2)	4	1
3	{a,c} (1)	1	0

False Positive / Hash Collision Solution
Bit Vector = [1, 0, 1, 0]

Collision Analysis:
• Bucket 2 is frequent (Total Count = $4 \geq 3$).
• The pairs in Bucket 2 are {a,b} (count 2) and {b,c} (count 2).
• Neither pair is frequent individually ($2 < 3$).
• Consequently, BOTH {a,b} and {b,c} pass into C2 as FALSE POSITIVES due to their hash collision into Bucket 2!